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Poster Session I. PM sampling and source apportionment

FIELD COMPARISON OF FOUR PM_{10} SAMPLERS IN A POLLUTED AREA IN THE NETHERLANDS

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INTRODUCTION

The new legislation from the European Community with the framework and daughter directives about ambient air quality contains strict limit values, among others, for fine particles (PM_{10}). In 2005, a yearly average of 40 $\mu g/m^3$ may not be exceeded, and this values is decreased to 20 $\mu g/m^3$ by 2010. The member states are obliged to monitor the concentration of PM_{10} at several sites and with an accuracy of 25%. There are a number of measuring instruments which are approved by the European Community and the EPA. The problem of measurement of suspended particles is that different instruments do not give the same results. In the Netherland, this leads to the situation that one approved instrument measures concentrations above the limit value and another instrument results in values below. An international field comparison has been performed by the European Community, but the results have not yet been published. From presentations of participants it is clear that there are local differences between correlation factors among instruments, which makes it difficult to come to a good interpretation of the measurements at other sites. This implies that a good evaluation of the air quality is still a problem. Aim of this study was to compare two frequently used automatic instruments for PM_{10} measurements (TEOM and ß-dust monitor) in a heavily polluted area in South Holland, in the surroundings of Rotterdam. As a reference, two filter samplers where added to the series, a Digitel and an Andersen with PM_{10} heads.

METHODS

The sampler comparison was performed in the period from october 1999 to april 2000 successively at three different sites in the city of Rotterdam: a site close to a motorway (Overschie), an industrial influenced site (Vlaardingen) and a site close to a trans-shipment area in the port of Rotterdam (Beerdam). At each site, the measurement was performed during at least four weeks. The following instruments for the measurement of fine suspended particulate matter were used: a TEOM 1400 (Tapered Element Oscillating Microbalans) from Rupprecht and Patashnick, a ß-dust monitor FAG FH 62 I-N, a Digitel DHA-80 and an Andersen Sierra 310 sampler, all equipped with PM_{10} heads suited for the flow speed of the instrument. The ß-dust monitor was from the National Institute of Public Health and Environment (RIVM) and was one of the instruments which are used in the National Air Quality Network in the Netherlands. The automatic instruments are continuous low-volume samplers, the Digitel is a medium-volume sampler and the Andersen a high-volume sampler. The two low-volume samplers both had an Andersen 236B PM_{10} head. The housing of TEOM and the ß-dust monitor were continuously held at 50 °C. The Digitel filter sampler had a slightly warmed inlet. But the filters were conditioned before weighing at 50 °C during 24 hours. This implies that for all instruments volatile compounds will not be measured. It is in this setup guaranteed that no water will remain on the sample. Differences in the results will be systematical differences in the instruments, and no differences caused by more or less evaporation of volatile compounds and water. Measurements of filters which have been dried at 20 °C and 50% RH might still contain water associated to the particles and could thus result in too high values.

RESULTS

From the first results at the industrial site Vlaardingen and at Beerdam it is clear that the TEOM measures lower concentrations than the ß-dust monitor by about 30-40% depending on the site. An exact correlation factor will be calculated at the end of the experiment. The reference filter methods show results which are systematically higher than the automatic methods.
The TEOM results were more stable and constant in time on a half-hour basis while the ß-dust monitor was much more variable on a short-time basis.
The results show that there are systematic differences between the different instruments with EPA certification. More details about relations with temperature, wind speed or relative humidity will be discussed in the presentation.
The conclusion of this short comparison is that the differences between the samplers for PM_{10} are too high to come to a good assessment of the air quality with one instrument. A standard instrument which is used for all measurements in Europe would be a solution to come to a coherent picture of the concentration of fine particulate matter.